

ARUN JELSINGE

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SUMMARY

B.Tech Information Technology graduate with hands-on experience developing and testing software applications using Python, JavaScript, React, Node.js, FastAPI, and MongoDB. Experienced in building REST APIs, integrating databases, implementing authentication, debugging applications, and performing automated and end-to-end testing.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

Frontend: HTML5, CSS3, React.js, React 19

Backend: Node.js, Express.js, FastAPI, REST APIs

Databases: MongoDB, SQL, MySQL

Testing: Unit Testing, Integration Testing, API Testing, E2E Testing, pytest, Jest, React Testing Library

Tools: Git, GitHub, VS Code

Core: Data Structures, OOP, DBMS, Software Development, Debugging, Troubleshooting, API Integration

PROJECTS

RoadCare — AI Road Damage Detection Platform | January 2026 – April 2026

Python | FastAPI | React | MongoDB | OpenCV | YOLOv8 | H3

GitHub: github.com/jelsingearun/RoadCare---AI-Based-Road-Damage-Pothole-Detection-System

- Developed a full-stack road-damage application by integrating React, Python/FastAPI, and MongoDB, resulting in reporting, image-processing, validation, and authority-review workflows.
- Implemented application security by adding JWT authentication, role-based access, input validation, and rate limiting, resulting in controlled API access at 60 requests/minute.
- Validated application functionality by performing unit, integration, database, geospatial, API, and frontend E2E testing, resulting in coverage across 6 testing categories.

Academix — Academic Collaboration Platform | July 2024 – November 2024

React 19 | JavaScript | Node.js | Express.js | MongoDB | TensorFlow.js | JWT

GitHub: github.com/jelsingearun/Academix

- Developed a full-stack collaboration platform by combining React 19, JavaScript, Node.js, Express.js, and MongoDB, resulting in integrated student collaboration workflows.
- Built backend functionality by implementing REST APIs, authentication, input validation, persistence, and role-based access, resulting in structured application operations.
- Integrated peer-matching functionality by applying TensorFlow.js scoring and ranking, resulting in compatibility-based student matching using available profile and activity data.

GitHub Repo Automator | January 2026

Python | Git | GitHub CLI | Security Automation

GitHub: github.com/jelsingearun/github-repo-automator

- Created a repository automation CLI by supporting folders, ZIP archives, and individual files, resulting in automated GitHub repository creation across 3 input formats.
- Implemented security scanning by detecting credentials, private keys, suspicious functions, and high-entropy secrets, resulting in coverage across 6 threat categories.
- Automated repository setup by handling Git initialization, repository creation, secret sanitization, templates, logging, and dry-run execution, resulting in validation across 5 functional areas.

EDUCATION

Malla Reddy University, Hyderabad

B.Tech in Information Technology | 2022 – 2026 | CGPA: 7.7/10

LEADERSHIP

Class Representative — Malla Reddy University | 2024 – 2026

- Represented 120+ IT students by communicating academic requirements, announcements, and concerns with faculty, resulting in structured student-faculty coordination.
- Coordinated 10+ technical workshops and academic activities by managing communication and event requirements, resulting in organized execution of academic initiatives.

CERTIFICATIONS

- HackerRank: SQL (Basic)
- IBM Skills Network: Introduction to Python Programming
- NPTEL: Cloud Computing